

IC 152: Introduction to Python and Data Science

Course Code	IC 152
Course Name	Introduction to Python and Data Science
L-T-P-C	3-0-2-4
Prerequisites	Institute core/consent of faculty member
Intended for	B.Tech
Distribution	Institute Core
Semester	Odd
Approval	25th BoA (Jul 2018); 17th Senate (Aug 2018)
Revised	47th BoA (Sep 2022) and 37th Senate (Oct 2022) - renamed from "Computing and Data Science"

Course Objectives

The first course in computing is a core course to all engineering disciplines as the use of computers in every engineering discipline requires mastering efficient computing for problem solving. Computers are used for mathematical analysis, data analysis, numerical problem solving, communications, and for many specialized applications.

Many engineering applications being are becoming largely data driven. Inexpensive sensors can be found in engines, mobile phones, home appliances and more elaborate ones in various measuring equipment like seismographs or medical equipment. Large corporations collect large amounts of user data from various online frameworks (like social media, emails etc.). Many businesses offer customized services to their customers based on data analysis. In this regard, an exposure to the science of data analysis would permit students to appreciate this paradigm.

This course is the first on a three-course series on computation, data science, statistics and introductory machine learning.

Course Contents

- **Introduction:** History, basic structure of a computer, network of computers (3 lec)
- **Program:** Algorithms + Data, programs as a sequence of instructions. Levels of programming: high-level language, medium-level language, machine language, interpreted and compiled languages (2 lec)
- **Memory:** variables, types, scalar, composite types, arrays/lists, expressions (3 lec)
- **Functions:** scope of variables, code reuse, call frame and call stack, arguments, return values (3 lec)
- **Control flow:** decisions, iterations (3 lec)
- **Data structures and objects:** lists, dictionaries, NumPy arrays, strings (8 lec)
- **File IO, command line arguments** (3 lec)
- **Data visualisation:** plotting functions, making use of colour, geographical data (4 lec)
- **Program development:** Testing, test cases, debugging, program efficiency (3 lec)
- **Case studies:** Modeling using statistics, curve fitting, interpolation, histograms, classification, correlation (6 lec)
- **Quizzes, review etc.** (3 lec)

Text Books

1. Michael Dawson, *Python Programming for the Absolute Beginner*, Third Edition, Course Technology PTR (Chapter 1-7)
2. *The Python Workbook: A Brief Introduction with Exercises and Solutions*, 2014 Edition
3. Jake Vanderplas, *Python Data Science Handbook* (Chapters 1-4)
4. R. G. Dromey, *How to Solve it by Computer*, Pearson, 1982
5. V. A. Spraul, *Think Like a Programmer: An Introduction to Creative Problem Solving*, No Starch Press, 2012 (Soft-copy available from the Library website)

References

1. Allen Downey, Jeffrey Elkner, Chris Meyers, *How to Think Like a Computer Scientist: Learning with Python*, Green Tea Press (Chapter 1-9, Chapter 11)
2. Jose Unpingco, *Python for Probability, Statistics and Machine Learning*, Springer.